MILITARY SPECIFICATION

CHARGER, BATTERY: 10541243

This amendment forms a part of Military Specification MIL-C-60299A(MU), dated 18 June 1970.

Paragraph 1.1: Delete and substitute:

"1.1 This specification covers an electronic battery charger which is operated from a direct current source. The charger is a system component for the Range Finder, LASER, AN/GVS-3 (see 6.1)."

Paragraph 3.5.1 and 3.5.1a: Delete and substitute:

"3.5.1 Charger and power supply.- With 18 to 32 VDC applied to J4 the battery charger assembly shall:

a. Indicate by "POWER ON" lamp when the power switch is set to the "ON" position."

* Paragraph 3.5.1b, line 2: Change "475 ma ± 10%" to "490 ma ± 15%".

* Paragraph 3.5.2, line 2: Change "475 ma" to "490 ma".

* Paragraph 3.5.2, Table I: Delete and substitute:

"TABLE I

Shut off voltage (Volts) ± 10%	Temperature (°F)	Resistor ohms ±1%
12.5	-25	1615
10.5	+77	870
10.0	+130	718

Paragraph 3.6: Change Drawing Number "10544747" to "10544752".

Paragraph 4.3.2.1, last sentence: Add Table "V".

* Paragraph 4.4.2, Table IV, delete in its' entirety and substitute the following:

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"TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	INSPECTION METHOD
304.	Vibration	3.4.1	4.6.1.4
305.	Shock	3.4.2	4.6.1.3
306.	Storage temperature	3.4.4	4.6.1.1
307.	Immersion	3.4.5	4.6.1.7
308.	Operating temperature	3.5	4.6.1.2"

Paragraph 4.4.2.1 as follows:

"4.4.2.1 Three (3) battery chargers shall be selected at random as a special sampling at 25%, 50% and 75% completion of the contract production. Each sample shall be subjected to the tests in Table V and shall then meet the requirements and tests in Table II."

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	INSPECTION METHOD
309.	Humidity, operating	3.4.3.1	4.6.1.5
310.	Humidity, non-operating	3.4.3.2	4.6.1.6
311.	Sand and dust	3.4.6	4.6.1.8
312.	Salt spray	3.4.8	4.6.1.10
313.	Electromagnetic compatibility	3.7	4.6.4

Paragraph 4.5.3, item 7: Change "75W" resistor to "175W".

Paragraph 4.6.1, Delete and substitute:

"4.6.1 Environmental.- The charger shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. The charger shall be subjected to performance tests specified in 4.6.2 through 4.6.4 at ambient conditions upon completion of all environmental tests unless otherwise specified in 4.6.1.1 through 4.6.5. Lid of charger shall be in the open position for humidity, sand and dust, fungus and salt spray tests."

Paragraph 4.6.1.2 Last sentence, delete and substitute:

"At the end of each 2 hour period, and while still at the specified temperature, the charger shall be tested in accordance with 4.6.2 to the applicable performance requirements of 3.5."

Paragraph 4.6.1.5, fourth line:

"Change "4.6.2 through 4.6.4" to "4.6.2 through 4.6.3".

Paragraph 4.6.1.5, add NOTE as follows:

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"NOTE: Excess moisture shall be removed from exterior surfaces of unprotected plugs prior to conduct of performance tests."

Paragraph 4.6.1.6, second sentence: Delete and substitute:

"Performance tests of 4.6.2 through 4.6.3 shall be initiated within one (1) hour and completed within three (3) hours of the end of the humidity cycle."

Paragraph 4.6.1.9, add NOTE as follows:

"NOTE: Battery charger subjected to fungus test is not required to be exposed to the environments of Table IV or V."

Change paragraph number "4.6.1.6" to "4.6.1.10".

Paragraph 4.6.2: Delete and substitute:

"4.6.2 Performance.- Conduct the tests of 4.6.2.1 through 4.6.2.4 with input voltages of 18.0 + 0.1 volts and 32.0 ± 0.1 volts applied to J4 (Pin A (+), Pin B (-)."

Paragraph 4.6.2.1: Delete and substitute:

"4.6.2.1 Power on indicator.- Set the power switch to the "ON" position. The indicator lamp located behind the switch shall light. Set the switch to the off position. The lamp shall not be lighted."

Paragraph 4.6.2.2, second sentence: Delete "DC" and substitute "ON"; delete last sentence.

* Paragraph 4.6.2.2, line 5: Change "475 ma ± 10%" to "490 ma ± 15%".

Paragraph 4.6.2.2, line 8: Change "3.6.2" to "3.5.2".

Paragraph 4.6.2.3, third sentence: Delete and substitute "Set the power switch to the ON position"; delete last sentence.

Paragraph 4.6.2.3, 4th sentence: Delete and substitute:

"Observe that transient component conditions shown on oscilloscope are not in excess of 1.5V peak to peak, and voltmeter readings of 8.5 ± 0.1 Vdc on V_1, and 26.5 ± 4.5 Vdc on V_2."

Paragraph 4.6.2.3, 6th sentence: Delete and substitute: "Observe that transient component conditions and voltmeter readings remain within specified limits."

Paragraph 4.6.2.4, fourth sentence: Delete and substitute "Set the power switch to the ON position."

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Paragraph 4.6.2.4, 7th line, change: "18.0 ± 01V" to "18.0 ± 0.1V".

Paragraph 4.6.2.4, 8th line, change: "28.0 ± 0.1V" to "32.0 ± 0.1V".

* Paragraph 4.6.2.4, 9th line, change: "10 amperes" to "10.5 amperes".

Paragraph 4.6.5, delete in its entirety and substitute:

"4.6.5 Reliability.- Reliability testing is required to be performed in conjunction with First Article testing. The reliability sample shall consist of six (6) chargers and a minimum of 400 hours shall be accumulated on each charger during reliability testing for requirements of 3.8. The battery charger shall be operated at standard ambient temperature (75 ± 5°F) after exposure to the non-operating environments (humidity, non-operating, sand and dust, and salt spray excluded) and during exposure to the operating environments of Table IV and V. Aminimum of 24 hours shall be accumulated with the BC operating continuously and simultaneously in the charging and ranging mode after/during each environmental exposure. (Humidity operating, 12 hours in lieu of 24 hours).

a. Charging mode.- The battery charger shall be connected as specified in 4.6.2.2 with input voltage of 18.0 + 0.1 volts. The variable resistances shall be set a 0 ohm. Each ammeter shall read 475 ma ± 10% and each volt-meter shall read 0 volts. This condition shall be maintained for a three (3) hour period. At the end of the three (3) hour period, the resistances shall be increased until the voltmeters attain the value specified in 3.5.2. The ammeter readings shall be reduced below 10 ma. When all three (3) ammeter readings are reduced below 10 ma, observe that the indicator lamp shall light. The above shall be repeated using 32 ± 0.1 volts in lieu of 18 + 0.1 volts. This constitutes one cycle. The above procedure applies to all three (3) charging stations and shall be repeated until 24 hours minimum operation is accumulated on each station. (12 hours during humidity operating test.)

b. Ranging mode.- The battery charger shall be connected as specified in 4.6.2.3 with input voltage of 18 + 0.1 volts. The variable resistances shall be maintained at 0 ohm for a three (3) hour period. The voltmeter shall indicate 8.5 ± 0.1 Vdc on V_1 and 26.5 ± 4.5 Vdc on V_2. The variable resistor shall be adjusted to 58 ohm and 294 ohms as specified in 4.6.2.3 and maintained for three (3) hours. The output voltage shall remain within the specified limits. The above procedure shall be repeated using 32 ± 0.1 V in lieu of 18 + 0.1 volts. This constitutes one cycle. The above cycle shall be repeated until the 24 hour minimum operation is accumulated (12 hours during humidity). At the conclusion of the environmental exposure the battery charger shall be conditioned at standard room temperature 70 ± 5°F and operated as above (charging and ranging mode) until the following relationship

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is satisfied:

$$850 \leq \frac{2T}{\chi^2_{\alpha,(2r+2)}}$$

$$\chi .05,\ 2(r+1)$$

Where T = Operating time

$\chi^2_{.05,\ 2(r+1)}$ = Chi square value lower 95% confidence level with 2 (r + 1) degrees of freedom

r = Number of failures.

Data collected shall include total operating time, time each failure occurred, type and source of failure, environmental exposure, time to correct failure, and action taken to allow resumption of testing.

A failure is defined as any malfunction which renders the battery charger inoperable or operable outside the limits of this specification.

Figure 1: Delete lines connected to pin "D"; connect lines to pin "F".

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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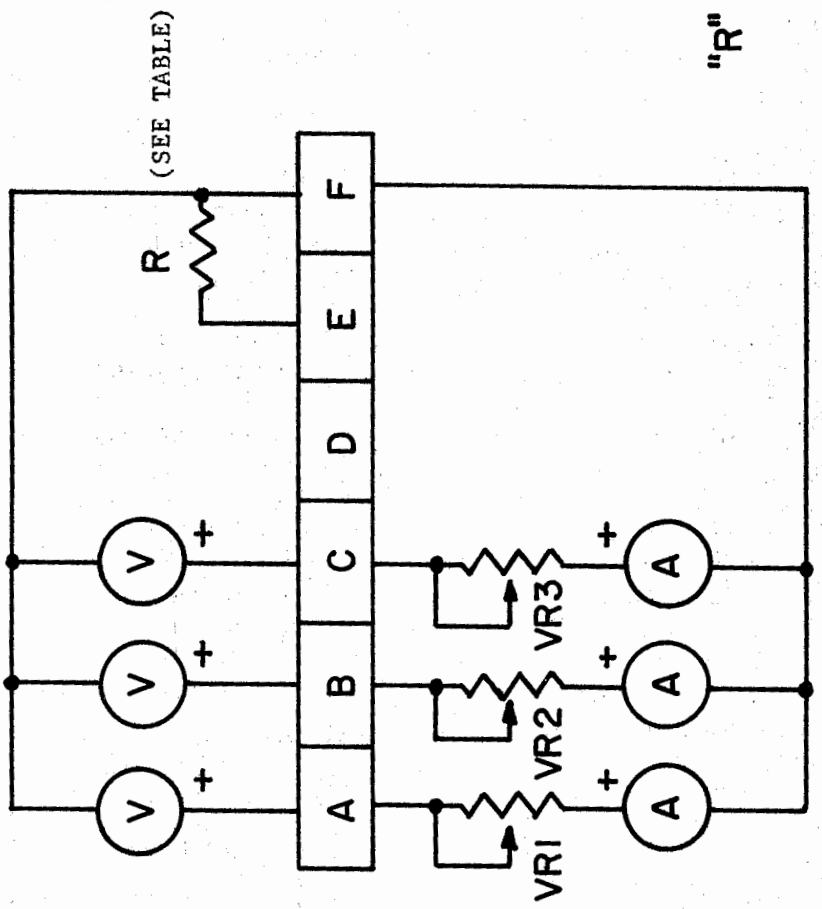
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"R" 870 ohms ±1% @ 77°F
718 ohms ±1% @ 130°F
1615 ohms ±1% @ −25°F

TEST CIRCUIT, BATTERY CHARGING OUTPUTS

FIG. 1

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